SEMTM0010 Transport and Mobility Modelling
2026 / 2027 coursework

General points.

- This coursework is open-book, and you are permitted to consult any material (e.g., lecture slides, exercises) that has been posted on the unit's Blackboard page, or indeed your own notes. You are also strongly encouraged to research on the internet, however, significant sources should be cited in an appropriate manner.

- Note this is an individual assessment, and your answers must be your own, and may not be obtained with any outside help. Collaboration is not permitted. If cheating or collaboration are detected, as with any assessment, then this will be subject to the University's rules on academic integrity:
  http://www.bristol.ac.uk/students/study/teaching/integrity/.

- Please take note of the University's rules on the use of AI. The default level permitted for this assignment, unless stated otherwise, is **category 2: minimal** — which means for example, it is ok to use AI to learn, and it is ok to use spelling and grammar checkers to help identify mistakes, but not to rewrite chunks of text. Furthermore, note at level 2, the use of AI to generate submitted computer code is not permitted.

- Some components of this coursework have been designed to be solved by AI, that is **category 4**. This is signalled in the relevant question components where further instructions are given. Note that for such components, marks will focus on how you explain the process of how you interacted with the AI to develop your solution and assure its quality, rather than on the quality of the AI output per se. For example, you might provide prompt sequences as evidence.

- The coursework consists of six compulsory questions, Q1–6, one for each section of the course. Each question is worth 20 marks. Each question is annotated with its mark breakdown. All six questions should be attempted.

- The computing elements can use Python, Matlab, or other languages to solve problems. The key issue is that code should be structured and explained so that it can be understood by a programmer who is non-expert in the particular language.

- Your submission should constitute a single pdf and might comprise a blend of (for example), pages of (neat) hand-written calculation or argument, figures — hand-drawn or powerpoint/similar, typed text, commented computer code, computer output (please label and annotate all graphs), and AI prompt interactions. There are many ways of addressing this. For example, the use of Matlab livescript or Jupyter in Python is a really neat way of combining (latex) documentation and code. However, livescript/Jupyter/latex are not compulsory and any method which gives a clear and easy-to-follow narrative is fine. However — we must emphasise: your submission should be bound together as a single pdf.

- You may ask questions about this coursework on the associated Blackboard discussion forum, with a cut-off of Friday 20 November.

- Your coursework should be submitted via Blackboard, with a deadline of 13:00 GMT on Tuesday 8 December 2026.

Q1. Demand Modelling.

Q2. Network modelling.

turn over/...

Q3. Assignment modelling.

Q4. Microscopic modelling.

This question is based on the following video on Blackboard: **crowd_documentary.mp4**. You need to watch the whole video before attempting the question. Please do not share the video due to copyright reasons. Where needed, use the fundamental diagram

$$v(\rho) = v_0 \left(1 - \frac{\rho}{\rho_{\max}}\right)^2, \qquad \Theta(\rho) = \rho\, v(\rho), \tag{1}$$

where Θ [m^{-1}s^{-1}] is the specific flow, ρ [m^{-2}] is the crowd density, v [m/s] is the average walking speed, v_0 [m/s] is the free speed, and ρ_{max} [m^{-2}] is the maximum crowd density.

(a) The video describes how a critical building at the Hajj was redesigned from one storey to several storeys.

(i) *Measure the crowd.* Pause the video and use one or several freeze frames that show pedestrians walking at ground level: from around the time *18:15*. For your chosen frame(s), estimate the crowd density (with an uncertainty range). Please include annotated frame(s) for reference.

(3 marks)

(ii) *Assess the infrastructure.* Use the annual number of pilgrims quoted in the video, and assume 80% of them perform the ritual within one 3-hour window. Each storey provides a walkway of effective width $W = 40\,\text{m}$. Use Eq. (1) with $v_0 = 1.2\,\text{m/s}$ and $\rho_{\max} = 7\,\text{m}^{-2}$. Take your crowd density from (a)(i) as the operating density on every storey. Compute the required minimum number of storeys, and compare it with the number of storeys mentioned in the video. Critically analyse how sensitive the infrastructure requirements are to the crowd density around your estimated range.

(3 marks)

(b) *Beyond the Hajj.* The video covers many incidents and situations around the world. State three specific examples from scenarios other than the Hajj in the video that relate to competition for scarce resources or to crowd psychology. For each, give the time in the video, and explain how it can (or cannot) provide further insights that may help with your Hajj problem. For full marks, please make reference to both audio and visual informaton in the video.

(4 marks)

(c) Unlike the presenter in the video, you have technical knowledge about microscopic crowd simulation.

- Based on the original social force model from 1995, provide all model equations (define every symbol), and propose any extension(s) required in order to simulate the scenario in (a), and incorporating concepts from (b). Please justify your chosen model extensions.
- Provide a plan for how your model can be run, which metrics or insights will be extracted from the model runs, and explain how those metrics or insights can be used to improve crowd safety at the critical building introduced in (a).
- For full marks: implement the model outlined above in a language of your chosing. Feel free to base it on an open-source implementation of the social force model. Perform the model runs, collect the output or metric data, and critically analyse this data.

(10 marks)

Q5. Data-driven modelling

For all parts of the question, please provide your answers as bullet points rather than continuous text. Your answer to this question should not be longer than 5 pages A4. Answers should be supported by appropriate plots showing results.

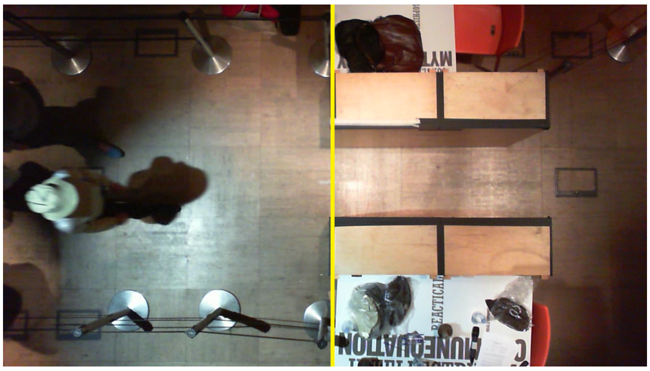

Figure 1: Screenshot of turnstile experiment.

This questions is concerned with a transport authority wanting to assess how many people can pass through opened turnstiles depending on passenger profiles. Specifically, the difference between crowds composed of families versus crowds composed of individuals (e.g. commuters) is of interest. As part of the investigation, experiments with volunteers were conducted. Consider the two videos families.mp4 and individuals.mp4 available on Blackboard. They show the footage from the experiments with the former showing a crowd of simulated families of 3 to 4 people (as indicated by numbers on hats) and the latter showing a crowd of individuals (ignore numbers on hats). A screenshot of the start of one of the videos is shown above. In the experiments, pedestrians are defined to have entered the turnstile if the centre of their head has passed the yellow line shown in the screenshot in figure 1 (this line is not shown in videos). The transport authority is interested in the time interval between consecutive people entering the turnstile, as the flow can be estimated from the inverse of these intervals. For example, if person A enters at time $t_1 = 1.6$ seconds, and person B enters at $t_2 = 2.4$ seconds, then the time interval is 0.8 seconds and starts at t_1.

Answer the following questions:

(a) Design, justify, and conduct data collection from the videos, recording the time intervals between consecutive people entering the turnstiles. Your answers to the later parts of this question will depend on your decisions at this point in the question, so we recommend reading the remainder of the question carefully.

You are permitted to use any method for obtaining the data (e.g. manual or automated). Deviation above 20% for time intervals and 3% for total interval counts from the ground truth will lead to marks deduction for results that is carried forward in the question up to a total of 10 marks. There is no requirement to extract all possible data from the videos but you must justify your data collection strategy, and describe how you obtained the data from the videos. If you are using an automated technique for extracting time intervals from the videos, do not include your code for this in your answer. Do include tables for each video

Qu. continues ...

with time intervals and their start times in the video that you obtain from your data collection (the tables are not included in the page count).

(10 marks)

(b) Fit time series models to the data you have extracted from the videos and show the code for this. Only consider model types discussed in lectures or labs. Justify your choice of model, discuss the model fit and what it tells you about differences between the two videos.

(6 marks)

(c) Critically discuss, with reference to your analysis from question 5(b) and any additional evidence, the usefulness of the experiment to answer the transport authority's question on the effect of audience profile on the flow rate through opened turnstiles.

(4 marks)

We remind you of the university policy that assessment materials, the videos in particular, must not be shared with anyone or anywhere outside the university.

Q6. Practitioner studies.

For all parts of the question, please provide your answers as bullet points rather than continuous text. Your answer to this question should not be longer than 5 pages A4. Answers should be supported by appropriate plots showing results.

Consider the Ada Lovelace Building, Tankard's Close, Bristol, BS8 1TW, on the precinct of the University of Bristol.

(a) Construct a network of the evacuation routes for the ground (reception level) and first floors of this building. The nodes in this network should be places where building occupants might start their evacuation, or areas where different evacuation routes merge, or change (e.g. a door). Edges in this network should be segments of evacuation routes where building occupants can move without other routes joining or without route properties changing, such as a corridor. For nodes where building occupants may start their evacuation, provide estimates of maximal occupancy. For edges, provide details relevant to the time it will take people to pass through them. There is no expectation that every detail of the building is considered when constructing the network, it can be simplified. Your simplifications and their assumptions, as well as how data to construct the network was obtained should be explained and justified. Please read the remainder of the question before attempting this part.

(10 marks)

(b) Design, develop, explain and exhibit code to simulate an evacuation of the ground and first floors of the Ada Lovelace Building at an estimated maximal occupancy on your network using your adaptation of the model described in

`https://doi.org/10.17815/CD.2020.24`

This simulation will also make simplifying assumptions that should be discussed critically. It is sufficient to only include the clearly-annotated code for the main event-driven simulation in your answer.

(6 marks)

(c) Building on your answers to parts (a) and (b), provide an estimate for the Required Safe Egress Time (RSET), the total time it takes all occupants to safely exit the ground and first floors of the Ada Lovelace Building at an estimated maximal occupancy, from the moment the fire alarm sounds.

(4 marks)

Guideline to allocation of marks for part:

- These are open-ended questions, there is not one correct answer. You are being marked on the level of knowledge, application, and critical appraisal of the methodological background.
- Excellent answers will go beyond what is explicitly taught on the course.